

Systems management of operational support systems applications

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This paper discusses some of the challenges of providing systems management to BT's operational support systems (OSS). As the OSS evolve to exploit the three-tier distributed architecture, the provision of successful systems management solutions must overcome many problems, including those of scale, scope, functionality, integration, rate of change, and the relative immaturity of available solutions. The enterprise systems management framework (ESMF), BT's solution to the systems management requirements of three-tier distributed computing systems, is described. The ESMF takes a standards-based approach, integrating suitable tools into a management framework. This framework has then been extended in novel ways to provide the overall solution.

1. Introduction

Successful management of BT's operational support systems (OSS) is an important part of supporting BT's business operations. Without it, these systems would not be fully able to play their crucial role in maintaining BT's service.

The discipline for providing this is 'systems management'. This embraces management of applications and services as well as systems.

- Systems are the combination of computing hardware and software that make a platform on which applications run. Systems are typically independent of business function, although they are often tailored to provide support for a specific set of applications. In the distributed environment, the network hardware and software used to interconnect applications and systems may also be viewed as systems.
- Applications are the sets of programs and data that provide function to the business. Each application runs on a suitable system.
- Services are what the end user sees and uses. Each service is an integrated set of systems and applications, supported by management processes.

The challenge of managing the OSS increases in difficulty as the systems move from one-tier to three-tier distributed computing. The requirement increases from management of just the systems to management of the services. BT has responded to this need for increased systems management with the enterprise systems management framework (ESMF).

2. The move to distributed computing

Distributed computing can be said to occur whenever two applications co-operate over a network to provide a business function. There are several different styles of distributed computing, including:

- client/server, where a user-driven application makes requests of a central service,
- pipeline, where, for example, batches of billing data are processed by pricing engines on their way to a billing system,
- groupware, where messages are stored and forwarded between systems.

Each of these has different ways of providing the total service. For example, groupware messages can be delayed by an hour and the users may still be satisfied. However, if a client/server transaction is delayed by a few seconds, the users will, quite rightly, complain. The management of these systems must reflect these differing requirements.

There are many reasons for moving to distributed computing. For example, client/server services allow clients to be created which empower people to do their jobs fully, while maintaining central control over the company's key business data. The reasons behind BT's shift to distributed computing are described elsewhere [1, 2].

The scale of distributed computing within BT is very large by industry standards. It already numbers dozens of mainframes, hundreds of servers and thousands of PC

clients. Within two years, these numbers are expected to increase several-fold, reaching in the order of 1500 servers and 30 000 PC clients.

2.1 *Vendor solutions to systems management of distributed computing*

Distributed computing on this scale poses a challenge to most vendors of systems management tools. Vendors typically offer either LAN-oriented tools, which have problems scaling up to these numbers, or mainframe-oriented tools, which require too much software on each system.

The mainframe vendors also have problems of scalability — most organisations only have a few mainframes. Some vendors, typically those with a network management background, can handle the scale, but these vendors' tools tend to lack the richness of functionality that the others can provide.

2.2 *BT's framework solution*

The approach that BT has adopted is based around integrating components into a framework. BT has used standards to make the platform components interoperate with enterprise-scale management systems. By mixing and matching products using the same standards, it has been possible to create a systems management toolset that covers all platforms. This still needs to be extended to provide the functional richness that management users require.

2.3 *Systems management strategies*

Systems management was initially a manual activity, as it still is for many smaller systems. Automation started on large mainframes in the late 1980s. Within BT, this was a key component in enabling the consolidation programme, which produced a significant reduction in the number of computing centres required to provide BT's computing operations.

The trend to distributed systems became clear in the early 1990s, as exemplified by the approach taken on a number of OSS systems described elsewhere [3, 4]. A project was initiated to provide systems management for distributed systems. Initially called the open systems management framework (OSMF), because of its focus on open systems; this has now evolved into the enterprise systems management framework (ESMF), which embraces management across the three tiers of the distributed computing paradigm (mainframes, servers and PC clients).

There are some common themes across the mainframe and open systems environments, which have themselves evolved as the requirements have been refined and the technologies improved. These themes continue to apply to distributed computing, and include:

- reporting the problem to the operator — at first this meant delivering the events to the master console, avoiding operators having to actively look for problems, and has developed into the correlation of events, so that a single diagnostic report replaces multiple symptom reports,
- reducing the effort required from operators — initially this meant filtering out messages of no value, and has developed to mean delivering only messages which require operator action,
- managing many systems from one point — initially this meant displaying messages on a single screen, but is now developing towards dealing with many systems as a single system image,
- automating common tasks — this started with standard responses to operator questions, has developed to cover automated recovery from simple problems, and will develop further, through the automatic start-up and shutdown of systems, to the automatic implementation of changes based on configuration files,
- managing rather than operating — this is the combined effect of the other themes, 'operating' describing the practice of reacting to events which have occurred, a reaction that may not take place until the system has suffered a gross failure, and 'managing' describing the practice of actively looking for events and dealing with them as they occur, thereby avoiding the potential gross failure of the system.

3. **The challenge for systems management**

Distributed computing poses a number of challenges for systems management. These arise from the increases in scale and scope, the requirements for new functionality, the need to integrate systems management tools, the rate of change of applications, and the relative immaturity of the solutions available in some areas. Each of these factors would be a significant challenge in its own right; the combined effect of all of them is obviously an even greater challenge.

3.1 *Scale*

Within the one-tier computing environment, the mainframe systems and applications to be managed number in the tens or hundreds. Although each system may host many applications, these are often implicitly managed as a part of the system. By contrast, within the three-tier

computing environment, each user has a PC on their desktop which is a system in its own right, potentially in need of management — within BT, these desktop systems will ultimately number in the tens of thousands.

In addition, the middle tier contains hundreds, or more, of servers to be managed, and the PC clients may well require services from servers within the building on the local area network. At the same time, mainframe-centred networks, in which the mainframe-terminal cluster links can be monitored from the mainframe, are replaced by wide and local area networks (WANs and LANs), with a corresponding increase in the number of network elements to be monitored. Hence, there is a huge increase in the number of elements potentially in need of systems management.

3.2 *Scope*

While the increase in scale poses a major challenge in itself, simply managing the elements one-by-one does not provide a satisfactory solution. It is necessary to manage the set of elements which provide the end-to-end service, from mainframe to mid-tier server, to building server and to desktop PC client, over WANs and LANs, since the availability of all these elements is required to provide the service to the user. Providing usable representations of the end-to-end set of elements is a new problem to be solved.

By contrast, the systems management of one-tier computing has concentrated on the management of the systems as one task, and the management of the networks as another. This is a significant increase in the scope of systems management.

3.3 *New functionality*

The distributed nature of three-tier computing means that there are new functions to manage; several examples illustrate this.

- Given that the software to provide the business function is now spread between mainframe, server and desktop client, keeping the software synchronised across the three tiers is a significant new task. This will continue to be a challenge until software developers are able to develop software that allows interworking between different versions in the different tiers, thus removing the dependencies.
- To access their business applications, users may need to log on to multiple systems. Managing user accounts and passwords, keeping them in step across multiple platforms, becomes a major task. The responsibility for this can be passed back to the users, by requiring them

to log on multiple times with different user accounts and passwords on different systems. This causes problems when users are denied access because passwords have expired or incorrect passwords have been used. The solution here involves implementation of a corporate single sign-on regime, whereby one user account and password gives access to all the components that a user requires, but this is an area in which suitable solutions are still emerging.

- Some functions like scheduling and back-ups now need to be synchronised across the tiers. For example, work on one platform may need to run after the successful completion of work on another, or it may be necessary to ensure that a database on one system is backed up as soon as the database on another is successfully backed up, because recovery of one may require the other to be restored to the corresponding state.

3.4 *Integration*

Two approaches are possible when it is necessary to provide a function across a number of platforms — use a tool which is able to provide the function across all the platforms, or integrate the tools which provide the function on each platform. The better approach varies from function to function, but in a number of cases the second is preferred, because the best available tools can be specific to certain platforms. This introduces the challenge of integration, which includes both integrating the platform-specific tools with each other to provide cross-platform functionality, and integration of the tools into the framework, so that they can be managed through a common management user interface rather than through product-specific interfaces. This challenge may sometimes be compounded by the proprietary nature of some tools and interfaces.

3.5 *Rate of change*

One of the key justifications for the growth of three-tier distributed solutions is the need to allow rapid change to business functionality. It is the resulting separation of the presentation and business logic layers from the data management layer which leads to the three-tier architecture. The intention is to circumvent the applications development backlog, which BT, like most large organisations, faces. The inevitable result is an increase in the rate of change of applications; but this change now occurs across the much-increased number of systems in the distributed architecture, rather than on the limited number of mainframe systems. This is yet another source of increased systems management complexity.

At the same time, the technology used both to provide and manage the services is itself changing at an accelerating

rate. These changes are irresistible, as they enable more business function to be provided at less cost.

3.6 *Relative immaturity of solutions*

Finally, the available management tools differ in their maturity across platforms — in factors like the richness of their functionality, the robustness with which they provide it, and the support available. The most problematical tier in this respect is the desktop client. Unfortunately, of the computing tiers, this is the tier with the greatest number of elements to be managed.

4. **Systems management capabilities**

Systems management requirements are driven by the needs of both customers and operations. Within BT, tens of thousands of users are provided with computing services, and each of these users has to be provided with the correct set of services. These services use data that is essential for BT's processes, and this data must be well-managed, and provided only to the appropriate users. Systems must comply with regulatory requirements. Computing users within BT, like users everywhere, require their services to be available to the agreed schedules and to perform to the agreed targets, and expect the costs of these services to reduce over time.

For operations, the systems management functions have to support routine work within the environment of continual change, and manage that change, while supporting the availability and performance targets, and reducing costs. Day-to-day work must where possible be fully automated, with human intervention only when necessary to restore service to targets. However, software, hardware and people are not infallible, and routine work has to be supported by a housekeeping regime which ensures that it is quick and easy to restore service after a failure. This regime must also ensure that the data is available to the right people and secured from others. The change process needs to be well managed, and automated where possible, to minimise the effort required for each change. Records must be kept to demonstrate that BT is in control of its systems. User accounts must be administered, and their access to services and data carefully managed.

The functionality needed to satisfy these requirements can be grouped into a number of areas:

- monitoring systems, services and events,
- running regular work,
- supporting the housekeeping and recovery processes,
- administering user accounts and access,
- making changes,
- supporting manual processes.

4.1 *Monitoring systems, services and events*

The monitoring system needs to detect all positive and negative events within one or more systems, and to forward alerts to the appropriate systems management function. Positive events are things that will always be reported, such as the successful end of a job, or a hardware failure on a peripheral device. Negative events are things that should have happened, or should be happening, but which have not occurred. The monitoring system will generally have to be active in seeking these negative events, whereas it can be passive and needs only to listen for the positive events.

The monitoring system will be able to take action on receipt of events, to correct a possible fault, to find out further information, or to pass it on to another function. It will need to present important events to system operators. It should also be able to present summaries of the status of all components in the domain that it monitors, so that system operators can easily see where there may be problems.

There is also a need to monitor the quality of the service being delivered. Often the quality is measured in terms of availability to the customer, the response time the customer receives, the delivery of back-office functions (e.g. printed output) and the total cost of the process. A measurement system needs to be provided to cover these measures, and the many other variables which contribute them. The measurement process must be able to handle both immediate data (e.g. to find which component is causing a long response time) and historical data (e.g. to detect trends in the use of a system showing when it will fill up).

4.2 *Running regular work*

The system that runs work must be able to start work after a suitable set of events has occurred. It needs to understand the capabilities of the systems it controls, so that they are fully loaded, but not overloaded. It needs to understand the sequence of work required so that the work is done to time. It needs to create events if work is running late or other problems occur.

It is often an event that indicates that work should now be run. For this reason, the monitoring and running work systems are closely linked.

4.3 *Supporting the housekeeping and recovery processes*

Housekeeping covers a variety of regular tasks that ensure that systems continue to run smoothly. For example, all systems produce log files of their activity. If left unmanaged, these would fill all the available disk space, leaving no room to write new business data. In general,

housekeeping tasks involve tidying data files, removing duplicate data, and ensuring all is ready for the next day's work. When this work is enhanced with tasks to make effective use of disk space, this is known as storage management.

To recover from the loss of data, it must have previously been copied to a known, safe place. It can then be copied back to its operational position. Since the same sets of data will be copied frequently, it is essential to keep quality records of what data is where.

When a database system is used, the records must link to the database system's records. In some cases, copies of data are taken as it is changed, so that data can be recovered with minimal loss. Database systems often need specialised types of housekeeping, and these are supported by specialist tools.

4.4 *Administering user accounts and access*

Administration covers the setting up of user accounts, their passwords and their profiles (the set of functions, such as transactions, that a user is authorised to use), and the control of access to data (through profiles which control which users can access what data).

There should be a single system for administration, which should drive the security systems on each platform. This needs to be supported by tools to audit systems and handle reported security violations. Other tools are needed to set up standard profiles into which user accounts can be linked. At present, different systems have different methods for administering user accounts, although a few of them are linked. So administration is currently done by direct interaction with the security systems on each platform.

Users also experience problems, which need to be recorded and entered into the problem management system, so that they can be solved by the appropriate people.

4.5 *Making changes*

The change support tool should enable the automation of changes. Since many of BT's systems are replicated, these changes must be automated in a way that can be repeated. This will mean that local variables will need to be parameters to the change task. The values for these variables should be retrieved from a configuration management system. The change tools should handle the distribution of changes across platforms, providing any required synchronisation of install, activate and roll-back. These distribution and synchronisation functions are often found in software distribution tools.

The tools also need to record what happened on each change, and to provide this data to a central configuration management database. They need to prompt the configuration management system to run its own discovery routines. Some changes will be manual activities (e.g. adding new hardware). Their results still need to be recorded.

The configuration management system will record successive states of the various systems under its control, thus collecting together all the changes made in a change session. The change tool will need to record the individual changes and their results, but will not record the complete state. Thus configuration management and change systems are complementary.

4.6 *Supporting manual processes*

Tools are needed to support various manual processes within computing operations. These include processes such as problem management and change management. The tools need to support the organisation, which is itself distributed, with people performing different functions located around the country.

5. **The enterprise systems management framework**

BT's approach to automating the management of distributed systems has been based around Unix systems. This is partly because they are the common mid-level system between PC clients and mainframes, and partly because their great flexibility has allowed solutions to be created quickly and cheaply. Early distributed systems were nearly all Unix-based, so this focus on Unix was fully justified. This set of solutions was called the open systems management framework (OSMF). The OSMF has been extended to monitor applications running on other platforms, such as MVS mainframes and Windows NT clients and servers. It currently provides management, as opposed to monitoring, for open systems (Unix) platforms only.

BT is now deploying distributed systems that cover client PCs, Unix servers and MVS mainframes, and the OSMF alone is inadequate. However, the standards-based approach of the OSMF means that it can be used as the core of a wider system, the enterprise systems management framework (ESMF).

The ESMF itself is a complex distributed system, as it must have a presence across the entire distributed environment. There are four layers which together provide the ESMF service:

- NetView server,
- XSA (Extended Systems Administration) Manager,
- delegated manager,
- managed system software.

These are each described below, together with their roles, and the unique features which distinguish the ESMF from the products on which it is based.

It has been the goal of the OSMF and ESMF to use regular products wherever possible, and to comply with industry standards, allowing the programme to avoid vendor lock-in. This approach has largely been successful, with the underlying use of SNMP standards enabling the OSMF's scope to be extended, in the ESMF, to cover the whole enterprise.

6. NetView server

The ESMF has taken the IBM NetView for AIX product [5] and used it as the focal point for management of distributed services, in particular producing a new map application that is capable of displaying both host and service-oriented topologies of managed resources.

The NetView server is the access point for all management users, and provides the user interface. In order to support the user population, NetView servers are installed wherever there is a population of management users. Multiple NetView servers can participate in the management of a single domain of managed resources. User sessions are provided by the X11 Window System protocol across the local area network.

6.1 Topologies

In NetView, the role of the map application is to structure and manage views of managed resources. For example, the base product provides a map application that displays managed resources in an IP-based topology of networks, segments, nodes and interfaces. For systems management this view is inappropriate as it does not show the management user the resources for which they have responsibility, nor does it represent whether a given distributed system is providing service or not.

The ESMF map application is capable of showing resources in arbitrary topologies, and is used to present different kinds of management user with different views according to their responsibilities. For example, users responsible for systems on a site are shown those systems under a single site symbol; while those responsible for a single application across the globe are shown those systems under another symbol.

In addition, the ESMF implements the concept of the managed service component. These are objects which take some part in the operation of a service, and may be either real (such as a hard disk or an executing software process) or virtual (such as transaction rate). In either case, they are represented graphically and have their status managed by the ESMF map application. Service components can either relate to managed hosts or be distinct from any host. In the former case all service components related to a host will be represented in the map topology underneath the host symbol; in the latter, they will be represented elsewhere in the topology.

Once service components have been defined, it is then possible to classify their criticality to the service. For example, the database processes on each system may be critical to the service, whereas a disk which is used for archiving old data may not be. This criticality may be represented by configuring how the ESMF map application propagates the status of that component. For the database processes, any change in component status will be propagated immediately, so that it is visible at the higher levels of the topology. In the case of the archive disk, a failure of the component may only cause the host object to which it belongs to be considered marginal.

Having classified managed service components, a new topology based entirely on service components can be constructed. In this topology, the components from individual hosts are shown together in functional groupings. For example, a management user responsible for database service across a large number of hosts can see all database service components represented together — all changes in status that they see are directly relevant to their job. This contrasts with host-oriented views where the change in status of a host could be the responsibility of one of any number of management users. In addition, new rules for status propagation can be defined for the service topology, so that events that affect the entire service are visible at the highest level, not just one level up the hierarchy.

Complex service maps showing the logical connectivity between different services can also be specified. This is increasingly important in an environment where one service is dependent on another, and where it is important for management users to be able to see the effect on one service of a failure on another.

The ESMF therefore provides, in a single view, a graphical representation of the status of the service being provided. This is the most important view — it aligns the requirements of the end users of the OSS with those of the systems management users.

6.2 Management operations

In order to effect changes on managed components, the ESMF has integrated the XSA management operations into NetView so that management users can select one or more systems or service components, and directly affect those objects from a NetView pull-down menu. This allows management operations to be invoked without needing to know either the specific architecture of the remote system or, indeed, to have any prior knowledge of what system is to be affected. For example, if a service component representing a file-system changes status and needs attention, the management user can select that symbol and change either the file-system or the files on it without having to know the specific hostname. In a distributed system with a large number of managed systems, this is invaluable.

The ESMF supports a wide range of systems management operations covering the management of items such as user accounts, user groups, files, file-systems, processes, printers and paging space. In addition XSA acts as a key integration point for the management of distributed applications. Management of the application is as important as management of the systems on which the application runs. By integrating them with the ESMF, the tasks to start, stop, query and configure the application can be invoked remotely with a common user interface, audit facility and access control, and they can take effect across multiple instances of the application without the application developer having to be concerned with the detail of how this is achieved. For the simplest tasks, all that is required is the actual code to perform the operation to be delivered as part of the application and for the ESMF to have a new management operation registered with it.

The product currently supports this management functionality only on open systems platforms.

7. XSA manager

This central system provides all management database services, and is based upon the IBM Extended Systems Administration (XSA) Manager product [6]. As a central repository, it is remotely available to all managed machines and management servers using a proprietary access-controlled protocol running over TCP/IP. As well as detailed configuration information about managed machines (such as user accounts, file-systems and devices), it stores the audit trail for management operations and information about scheduled operations. All scheduled management operations are invoked from the XSA Manager to managed systems.

The ESMF has integrated new functionality into the manager to implement management user roles. A management role describes all attributes of the responsibilities that

a management user needs to perform their job — what events they should see, what remote operations are appropriate, what systems they should have access to, and so on. Each user is then assigned one or more management roles according to operational requirements. This approach allows complete control over management users at minimum management overhead.

The XSA database server has high availability requirements, and is implemented on a resilient system by using cluster management software across two systems which share a common resilient RAID disk sub-system, along with back-up to a remote site in case of disaster.

8. Delegated manager

In a large-scale deployment, such as that within BT, there is a requirement to off-load processing from NetView to delegated managers at local sites. It is the role of the delegated manager to make decisions about the events it sees on behalf of the NetView server, and report only those which are significant. This is particularly important in event processing and status polling.

The delegated manager's role in event processing is to reduce the number of duplicate event reports to a manageable level, and where possible to convert symptom reports, which merely report what has happened, into diagnostic reports, which report why it happened. For example, in a client/server system, failure of the server software will tend to lead to a failure event from each client, and a single event from the server. The delegated manager is tasked with restricting the number of events from the clients, and potentially sending only the event from the server. This is essential if management users are to investigate actual problems and not just symptoms reported from elsewhere.

Another task of the manager is to poll all the managed systems to ensure that they are network-reachable. In the ESMF, with a large number of NetView servers responsible for the same management domain, it is more efficient and reliable to pass this task to the delegated manager which can then report actual failures to the NetView servers.

Currently delegated managers have been implemented only for PC client-oriented business applications, where they are essential to deal with the scale involved. The IBM Systems Monitor product [7] has been used here in a limited way; previous versions have suffered from limitations which have restricted its applicability. A BT-designed delegated manager has been developed and is likely to be used until such time as a satisfactory external product is available.

9. Managed system software

ESMF has a presence on each managed system, providing the instrumentation and management functions that are needed to manage the system and its applications. This software, and its correct configuration, is vital to present the correct information to management users, and to permit the management operations to take place.

9.1 Event generation

At the lowest level, the ESMF provides an interface for the generation of SNMP traps [8–10] to NetView. SNMP traps provide a standard technology for the transmission of asynchronous alerts from managed systems to SNMP managers. A set of structured traps, registered under a subtree of the BT Enterprise to ensure global uniqueness, is used. Each trap allows for the transmission, in variables bound to the trap, of the information required to align the managed resource's and manager's understanding of the specific event.

This internal interface is then abstracted into a higher-layer function which allows integrators and developers to generate correctly formatted events with no knowledge of SNMP or its configuration. This allows many applications and pieces of systems software to be integrated into the ESMF by those people who best understand the function of that software, rather than by systems management specialists. It is also used by other management applications, such as resource managers, to generate appropriate events.

Other resources which need integrating into the ESMF do not provide for such an interface — instead events are written to log files. An ESMF mediation device is used to convert local log formats into appropriate traps. On open systems platforms this is capable of integrating log files, the system console and output from executed commands; on Windows NT it is capable of integrating the Event Log; on MVS it is capable of integrating NetView/AOC events.

As well as generating SNMP traps, systems are instrumented using SNMP agent products from appropriate vendors. These allow management users to perform *ad hoc* queries of the remote systems, supporting standard MIB-II and Host Resources (RFC 1514) or equivalent instrumentation where possible. It is also possible to further instrument either the system or the application by using extensible SNMP agents.

9.2 Management functionality

On open systems platforms, the ESMF build of the XSA software is installed. This provides support for the management operations, and can be extended to support management of the application.

As one of the ESMF's goals is to allow the integration of strategic management applications (such as back-up and restore, software distribution and security monitoring), each of these management applications will also be installed on the managed system. Each will have been integrated into the ESMF, so that they can be managed as any other application; in some cases they can be managed to a much more detailed level. For example, the back-up and restore service is integrated into the ESMF such that all operator actions are performed from the ESMF, and the back-up and restore product itself is never actually seen by operators.

10. The future direction of the ESMF

The future of the ESMF is divided into three phases, evolving towards the vision of the third phase. These phases are:

- to consolidate and exploit the existing technology,
- to widen the scope towards coverage of the enterprise,
- to enable true service management.

At each phase it will be necessary to review the position and priorities of the programme, ensuring that the solutions delivered are solving the problem of maximising computing service delivered to the customer at least cost to the business.

10.1 Consolidation

While the ESMF has been successful in its OSMF form, full exploitation of the current technologies needs further work. Provision of capabilities for larger numbers of management users, improved event correlation at both the NetView and delegated manager level, and wider roll-out of delegated management functionality, are all essential to consolidate the current service and prepare for a wider enterprise scope.

At the same time use of some enterprise-scope technologies, including Windows NT desktop client and MVS mainframe monitoring capabilities, will be required to support the new distributed applications.

During this phase there will also be further selection of strategic management tools, including examining how the desktop client can be more fully managed.

10.2 Enterprise scope

During this phase the ESMF will see the beginnings of a real enterprise scope, with technologies being deployed to present a more complete picture across desktop client, building server, network, open systems server, MVS mainframe, and other platforms. This will begin to allow the control of all such environments from a single display.

Implementation of such a direction will require new technology across all the parts. Just as SNMP has become all-pervasive, which is a strong reason for its use, the

industry trend is towards increasing commonality in the basic technologies and greater pervasiveness of those technologies across the range of platforms. This trend will support the evolution of ESMF in this phase.

10.3 Service management

During this phase the ESMF will move towards its vision of transparent control of service components without regard for the architecture on which they are built, with full integration of management functions across all layers. This will allow service managers, driven by the need to meet service level agreements but with few (if any) platform skills, to be able to control all aspects of the services for which they have responsibility.

The ESMF will then be a true framework stretching across all the platforms and systems upon which distributed systems are built. It will allow for the construction of business services which are transparently manageable regardless of the technologies used.

11. Conclusions

This paper has shown how the management of multiple distributed applications and the underlying components is itself complicated, and requires a complex distributed management application. The trend, both of applications and management, is towards increasing complexity. The technologies available to address this complexity have not been available until recently, and are still emerging and developing. Through the ESMF, BT is at the leading edge of understanding and resolving the problem.

The success of this approach will increasingly be dependent on support for systems management being designed into business applications. This will enable the ESMF to fulfil its key role in improving service for internal, and hence for external, customers.

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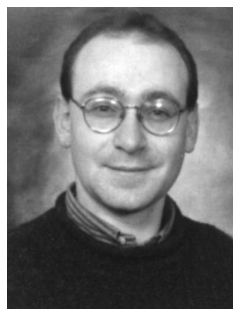
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